

# HECTOR SLAM MAPPING AND POSITIONING With ROS And Lidar

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In previous segments, we delved into lidar-based obstacle detection and the integration of lidar and lane detection systems for the ADAS system. If you are a design engineer or a newcomer, understanding the comprehensive design

of autonomous electric vehicles (EVs), drones, and ADAS systems is crucial. (For a comprehensive overview, refer to Part 1 and Part 2: Anti-Collision System articles published in the June and July issues.)

Building upon the foundation of the earlier ADAS system design, we

now progress to implementing lidar technology to scan the ROS environment. This scanning process generates real-time maps of the surroundings using simultaneous localisation and mapping (SLAM). Furthermore, our focus is on localising the vehicle within the map without relying on GPS data. Such a system within the ADAS and autonomous vehicle drive landscape enhances localisation accuracy and aids in path planning even in scenarios where GPS connectivity is compromised, such as tunnels or subways. It significantly aids in navigation by dead reckoning, which is a method of calculating an object's current position based on a



Fig. 1: Vehicle mounted with lidar

**TABLE 1**  
**BILL OF MATERIALS**

| Components                  | Quantity | Description         |
|-----------------------------|----------|---------------------|
| Nvidia Jetson/<br>Raspberry | 1        | 4GB RAM             |
| Lidar                       | 1        | 360 degree lidar    |
| SD card                     | 1        | 32/64GB             |
| HDMI display                | 1        | 17.8cm HDMI display |

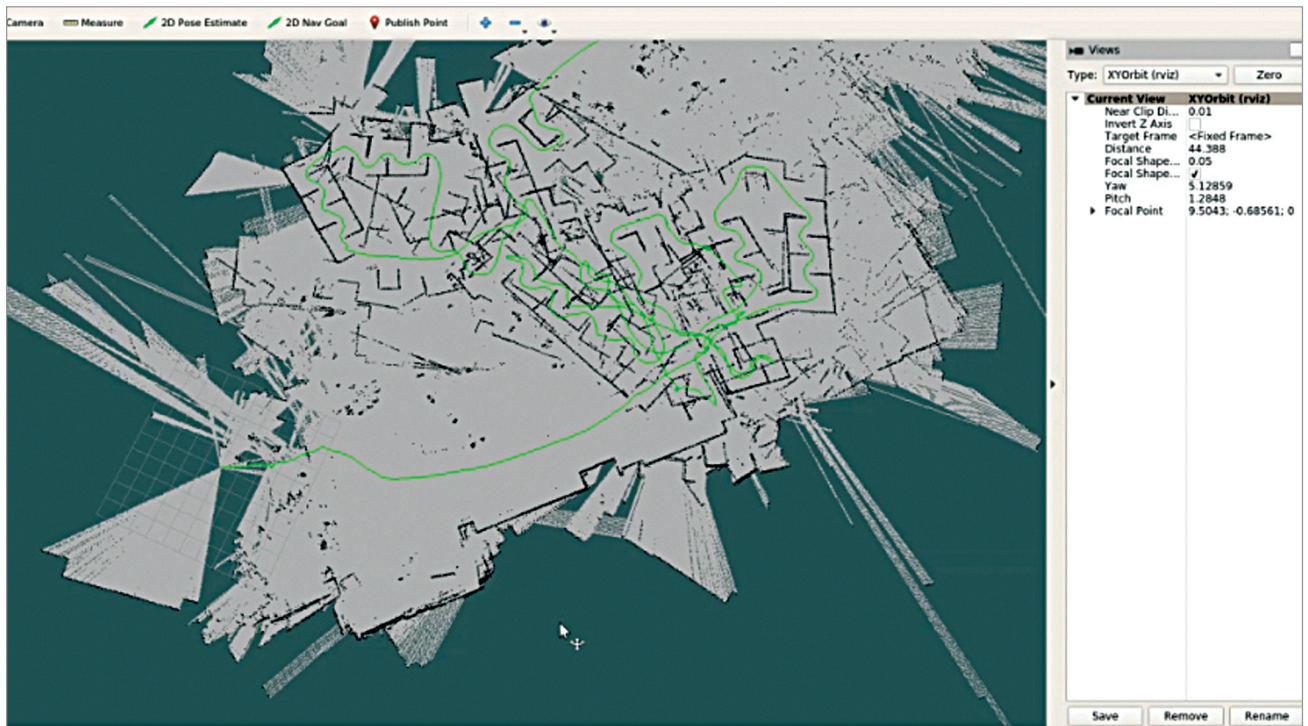


Fig. 2: Creating the path (green lines) on maps based on movement of vehicle